Chapter 1 – Introduction to EMC

1 Chapter 1: Introduction to Electromagnetic Compatibility

1.1 1.1 What is EMC?

Electromagnetic Compatibility (EMC) is about managing electromagnetic energy so that electronic systems coexist without causing interference. Three criteria must all be satisfied *(chap1.pdf, p.4)*:

1. The system **does not cause interference** with other systems.
2. The system **is not susceptible** to emissions from other systems.
3. The system **does not interfere with itself**.

Formal definition: *"An electronic system is said to be electromagnetically compatible with its environment if it does not produce or be susceptible to interference." (chap1.pdf, p.4)*

Note: coupling signal into a system is not by itself interference --- interference only occurs when the receptor behaves in an undesired manner. *(chap1.pdf, p.5)*

1.2 1.2 EMC Coupling Decomposition

EMC is concerned with three stages of electromagnetic energy: **generation** → **transmission** → **reception**. *(chap1.pdf, p.5)*

1.2.1 Ways to Prevent Interference (in order of ease/cost-effectiveness) *(chap1.pdf, p.6)*

1. Suppress the emission **at its source**.
2. Make the **coupling path** as inefficient as possible.
3. Make the **receptor** less susceptible.

Working from step 1 to step 3 is progressively harder and more expensive. The simplest (but most costly) fix is a fully enclosed metallic housing with internal batteries.

1.2.2 Four Basic EMC Subproblems *(chap1.pdf, p.7)*

#	Type	Description
1	Radiated emissions	Device radiates EM energy into the environment
2	Radiated susceptibility	Device is affected by external radiated fields
3	Conducted emissions	Noise passes out via the power cord
4	Conducted susceptibility	Device is affected by noise on the power cord

1.2.3 Power Net Concern *(chap1.pdf, p.8)*

The ac power cord is intended for 60 Hz, but higher-frequency signals inevitably exist on it. These can interfere with other systems via conduction or radiation --- hence power-net EMC is critical.

1.2.4 Other EMC Concerns *(chap1.pdf, p.9)*

- **ESD** (Electrostatic Discharge): e.g., static build-up reaching kV levels
- **EMP** (Electromagnetic Pulse): e.g., nuclear-generated burst
- **Lightning**: atmospheric discharge coupling
- **TEMPEST**: EM emissions that carry information (security concern)

These are not government-regulated in the same way but must be considered in design.

1.3 1.3 History and Regulation *(chap1.pdf, p.11)*

- **CISPR** (Comité International Spécial des Perturbations Radioélectriques, 1933): international committee under IEC; produced earliest EMI measurement-equipment standards.
- **FCC** (USA, 1979): required digital devices to limit EM emissions; governed by Part 15 of FCC Rules, Title 47 CFR.

1.4 1.4 Real-World Examples *(chap1.pdf, pp.12--14)*

Selected illustrative cases from the textbook:

- Motor spark-gap → power cord → TV interference
- Copying machine → power net → hall clock reset
- Surveillance radar → airport computer data loss
- Falkland War (1982): communication system interference → antimissile detection failure → destroyer lost
- UH-60 Black Hawk helicopter crashes due to EM emissions affecting electronic flight controls
- Ship radar → RF voltages → inadvertent missile deployment

These examples demonstrate that EMC failures span from annoyances to life-threatening situations.

1.5 1.5 Electrical Dimensions and Waves

1.5.1 Why Electrical Dimensions Matter *(chap1.pdf, p.15)*

The **electrical dimension** (size relative to wavelength) determines which model to use:

Condition	Model
$L \ll \lambda/10$	Lumped-circuit theory (KVL/KCL)
$L \sim \lambda$ or larger	Maxwell's equations required

1.5.2 Time Delay *(chap1.pdf, p.16)*

For a wave propagating from node a to node b along a conductor of length L:

$$T_D = \frac{L}{v}$$

(chap1.pdf, p.16)

where v is the phase velocity of the wave in the medium.

1.5.3 Phase Difference and Electrical Length *(chap1.pdf, p.17)*

A sinusoidal propagating wave traveling in the $+z$ direction:

$$V(z,t) = V^+ e^{j(\omega t - \beta z)}$$

(chap1.pdf, p.17)

where β is the phase constant and ω is the radian frequency.

For a wave traveling a length L, the **phase difference** between the two ends is:

$$\Delta\phi = \beta L$$

(chap1.pdf, p.17)

Since $\beta = 2\pi/\lambda$, the **electrical length** (phase difference expressed in wavelengths) is:

$$\Delta\phi = \frac{2\pi L}{\lambda}$$

(chap1.pdf, p.17)

1.5.4 Phase Velocity *(chap1.pdf, pp.18--20)*

Phase velocity relates angular frequency to phase constant:

$$v = \frac{\omega}{\beta}$$

(chap1.pdf, p.18)

The current can be written in time-delay form, showing that phase shift is equivalent to a time delay z/v:

$$I(z,t) = I^+ e^{j\omega(t - z/v)}$$

(chap1.pdf, p.19)

In a medium with relative permittivity ε_r and relative permeability μ_r:

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{v_0}{\sqrt{\mu_r\varepsilon_r}}$$

(chap1.pdf, p.20)

where the free-space velocity is:

$$v_0 = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

(chap1.pdf, p.20)

Key insight: wave velocity is **smaller** in denser media (larger ε_r). *(chap1.pdf, p.20)*

1.5.5 Wavelength and Electrical Dimension in a Denser Medium *(chap1.pdf, p.21)*

Wavelength in a medium:

$$\lambda = \frac{v}{f} = \frac{v_0}{f\sqrt{\mu_r\varepsilon_r}}$$

(chap1.pdf, p.21)

Electrical dimension of a circuit element of physical length L:

$$k = \frac{L}{\lambda}$$

(chap1.pdf, p.21)

In a denser medium (ε_r larger): v is smaller $\rightarrow$ λ is smaller $\rightarrow$ k is **larger** (more electrically significant). *(chap1.pdf, p.21)*

1.5.6 Reference Tables *(chap1.pdf, pp.22--24)*

- **Table (p.22):** Frequencies and corresponding free-space wavelengths for common electronic systems. Knowing the spectrum distribution helps identify interference sources.
- **Table (p.23):** Relative permittivities ε_r of various dielectrics.
- **Table (p.24):** Relative permeabilities μ_r and conductivities (relative to copper) of various metals.

1.6 1.6 Decibels and Common EMC Units

1.6.1 Power Gain *(chap1.pdf, p.25)*

Power gain defined as ratio of output to input power:

$$G_p = \frac{P_{out}}{P_{in}}$$

(chap1.pdf, p.25)

where the powers are:

$$P_{in} = \frac{v_{in}^2}{R_{in}}, \qquad P_{out} = \frac{v_{out}^2}{R_L}$$

(chap1.pdf, p.25)

Power gain in decibels:

$$G_{p,\text{dB}} = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

(chap1.pdf, p.25)

1.6.2 Voltage and Current Gains *(chap1.pdf, p.26)*

$$G_{V,\mathrm{dB}} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right), \qquad G_{I,\mathrm{dB}} = 20\log_{10}\left(\frac{I_{out}}{I_{in}}\right)$$

(chap1.pdf, p.26)

When $R_{in} = R_L$:

$$G_{V,\mathrm{dB}} = G_{I,\mathrm{dB}} = G_{p,\mathrm{dB}}$$

(chap1.pdf, p.26)

Factor-to-dB conversion table (p.27): multiplying a quantity by 2 adds 6 dB; by 10 adds 20 dB (voltage/current) or 10 dB (power).

1.6.3 Using dB for Amplifier Calculations *(chap1.pdf, p.28)*

Since $G_p = P_{out}/P_{in}$:

$$P_{out,\mathrm{dBm}} = G_{\mathrm{dB}} + P_{in,\mathrm{dBm}}$$

(chap1.pdf, p.28)

This also applies to dBμW and dBμV, dBμA (when $R_{in} = R_L$):

$$V_{out,\mathrm{dBmV}} = G_{\mathrm{dB}} + V_{in,\mathrm{dBmV}}$$

(chap1.pdf, p.28)

Gains and losses have **opposite signs**: a loss of x dB appears as $-x$ dB.

1.6.4 Power Loss in Cables *(chap1.pdf, pp.29--30)*

For a lossy transmission line, when the line is matched ($Z_L = Z_C$), the forward-traveling voltage wave is:

$$V(z) = V^+ e^{-\gamma z}$$

(chap1.pdf, p.29)

where $\gamma = \alpha + j\beta$ is the complex propagation constant and α is the attenuation constant (Np/m).

The average power delivered in the $+z$ direction:

$$P_{av}(z) = \frac{|V^+|^2}{2R_C}\, e^{-2\alpha z}$$

(chap1.pdf, p.29)

where $R_C = \mathrm{Re}(Z_C)$.

Power loss over cable length L:

$$P_L = \frac{P_{av}(L)}{P_{av}(0)} = e^{-2\alpha L}$$

(chap1.pdf, p.30)

Cable loss in dB:

$$P_{L,\mathrm{dB}} = 10\log_{10}(P_L) = 8.686\,\alpha L$$

(chap1.pdf, p.30)

Rearranging to find the attenuation constant from a measured cable loss (in dB per unit length):

$$\alpha\,[\mathrm{Np/m}] = \frac{P_L\,[\mathrm{dB/m}]}{8.686}$$

(chap1.pdf, p.30)

Important voltage--power relationship (valid when $R_L = 50\,\Omega$):

$$\mathrm{dB}\mu\mathrm{V}_{\mathrm{RMS}} = 107 + \mathrm{dBm}$$

(chap1.pdf, p.30)

1.6.5 Signal Source and Instrument Impedances *(chap1.pdf, pp.31--33)*

- Modern signal sources use a Thevenin equivalent with $R_S = 50\,\Omega$.
- **Spectrum analyzer:** $C_{in} = 0$, $R_{in} = 50\,\Omega$.
- **Oscilloscope:** $C_{in} = 47\,\text{pF}$, $R_{in} = 1\,\text{M}\Omega$.

Because the SA has $R_{in} = 50\,\Omega = Z_C$, the cable is matched, and the input impedance is $Z_{in} = 50\,\Omega$ regardless of cable length or frequency. *(chap1.pdf, p.32)*

This is why modern EMC test equipment uses $50\,\Omega$ throughout: a swept-frequency measurement is meaningless if the source output varies with frequency, which would happen with an unmatched load. *(chap1.pdf, p.33)*